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Constraints on Dynamical Dark Energy from Multiple Probes in the Full Dark Energy Survey

T. M. C. Abbott,[1] M. Adamow,[2] M. Aguena,[3,4] A. Alarcon,[5] S. Allam,[6] O. Alves,[7,8] A. Amon,[9] D. Anbajagane,[10] F. Andrade-Oliveira,[11] P. Armstrong,[12] S. Avila,[13] J. Beas-Gonzalez,[14] K. Bechtol,[14] M. R. Becker,[15] G. M. Bernstein,[16] E. Bertin,[17,18] J. Blazek,[19] S. Bocquet,[20] D. Brooks,[21] D. Brout,[22] D. L. Burke,[23,24] H. Camacho,[25,4] G. Camacho-Ciurana,[5] R. Camilleri,[26] G. Campailla,[27] A. Campos,[28,29] A. Carnero Rosell,[30,4,31] A. Carr,[26] J. Carretero,[32] F. J. Castander,[33,5] R. Cawthon,[34] K. C. Chan,[33,5] C. Chang,[35,10] R. Chen,[36] J.M. Coloma-Nadal,[37] C. Conselice,[38,39] M. Costanzi,[40,3,41] M. Crocce,[33,5] W. d'Assignies,[32] L. N. da Costa,[4] M. E. da Silva Pereira,[42] T. M. Davis,[26] J. De Vicente,[13] D. L. DePoy,[43] J. DeRose,[44] S. Desai,[45] H. T. Diehl,[6] S. Dodelson,[35,6,10] P. Doel,[21] C. Doux,[16,46] A. Drlica-Wagner,[35,6,10] T. F. Eifler,[47,48] J. Elvin-Poole,[49] S. Everett,[50] A. E. Evrard,[51,8] I. Ferrero,[52] A. Ferté,[24] B. Flaugher,[6] P. Fosalba,[33,5] D. Francis de Souza,[53] J. Frieman,[35,6,10] L. Galbany,[33,5] J. García-Bellido,[54] M. Gatti,[5,10] G. Giannini,[5,10] P. Giles,[55] K. Glazebrook,[56] D. Gruen,[20] R. A. Gruendl,[2,57] G. Gutierrez,[6] I. Harrison,[58] W. G. Hartley,[59] K. Herner,[6] S. R. Hinton,[26] D. L. Hollowood,[60] K. Honscheid,[61,62] E. M. Huff,[50,48] D. Huterer,[8] B. Jain,[16] D. J. James,[22] M. Jarvis,[16] N. Jeffrey,[21] T. Jeltema,[60] S. Kent,[6] R. Kessler,[35,10] A. Kovacs,[63,64] K. Koyama,[65] E. Krause,[66] R. Kron,[6,10] K. Kuehn,[67,68] O. Lahav,[21] J. Lee,[14] S. Lee,[69,48,50] E. Legnani,[32] T. S. Li,[70] A. R. Liddle,[71] C. Lidman,[72,12] H. Lin,[6] M.-X. Lin,[73,74,16] N. MacCrann,[75] J. L. Marshall,[43] S. Mau,[36] R. G. McMahon,[76,77] J. Mena-Fernández,[78,46] F. Menanteau,[2,57] R. Miquel,[79,32] J. J. Mohr,[20] J. Muir,[80] J. Myles,[9] A. Möller,[56] R. C. Nichol,[65] R. L. C. Ogando,[81,82] W. J. Percival,[49,83] D. Petravick,[2] A. Pieres,[4,82] A. A. Plazas Malagón,[23,24] B. Popovic,[36] A. Porredon,[13,84] J. Prat,[85] H. Qu,[16] M. Raveri,[27] J. Rebouças,[86] W. Riquelme,[87] M. Rodriguez-Monroy,[84,54,88] P. Rogozenski,[89] A. K. Romer,[55] A. Roodman,[23,24] R. Rosenfeld,[90,4] A. J. Ross,[61] E. S. Rykoff,[23,24] M. Sako,[16] S. Samuroff,[19,32] C. Sánchez,[91,32] E. Sanchez,[13] D. Sanchez Cid,[13,11] T. Schutt,[92,23,24] D. Scolnic,[36] I. Sevilla-Noarbe,[13] N. Shah,[65,21] P. Shah,[21] E. Sheldon,[25] M. Smith,[93] M. Soares-Santos,[11] E. Suchyta,[94] M. Sullivan,[95] M. E. C. Swanson,[2] B. O. Sánchez,[36,46] M. Tabbutt,[14] G. Tarle,[8] G. Taylor,[12] D. Thomas,[65] C. To,[35] L. Toribio San Cipriano,[13] M. Toy,[95] M. A. Troxel,[36] D. L. Tucker,[6] V. Vikram,[96] M. Vincenzi,[97] N. Weaverdyck,[98,44] J. Weller,[99,100] A. Whyley,[65] R.D. Wilkinson,[55] P. Wiseman,[95] M. Yamamoto,[9,36] B. Yanny,[6] B. Yin,[36] Y. Zhang,[101] and J. Zuntz[102]

(DES Collaboration)

[1]Cerro Tololo Inter-American Observatory, NSF's National Optical-Infrared Astronomy Research Laboratory, Casilla 603, La Serena, Chile
[2]Center for Astrophysical Surveys, National Center for Supercomputing Applications, 1205 West Clark St., Urbana, IL 61801, USA
[3]INAF-Osservatorio Astronomico di Trieste, via G. B. Tiepolo 11, I-34143 Trieste, Italy
[4]Laboratório Interinstitucional de e-Astronomia - LIneA, Av. Pastor Martin Luther King Jr, 126 Del Castilho, Nova América Offices, Torre 3000/sala 817 CEP: 20765-000, Brazil
[5]Institute of Space Sciences (ICE, CSIC), Campus UAB, Carrer de Can Magrans, s/n, 08193 Barcelona, Spain
[6]Fermi National Accelerator Laboratory, P. O. Box 500, Batavia, IL 60510, USA
[7]Kavli Institute for the Physics and Mathematics of the Universe (WPI), UTIAS, The University of Tokyo, Kashiwa, Chiba 277-8583, Japan
[8]Department of Physics, University of Michigan, Ann Arbor, MI 48109, USA
[9]Department of Astrophysical Sciences, Princeton University, Peyton Hall, Princeton, NJ 08544, USA
[10]Kavli Institute for Cosmological Physics, University of Chicago, Chicago, IL 60637, USA
[11]Physik-Institut, University of Zürich, Winterthurerstrasse 190, CH-8057 Zürich, Switzerland
[12]The Research School of Astronomy and Astrophysics, Australian National University, ACT 2601, Australia
[13]Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas (CIEMAT), Madrid, Spain
[14]Physics Department, 2320 Chamberlin Hall, University of Wisconsin-Madison, 1150 University Avenue Madison, WI 53706-1390
[15]Argonne National Laboratory, 9700 South Cass Avenue, Lemont, IL 60439, USA
[16]Department of Physics and Astronomy, University of Pennsylvania, Philadelphia, PA 19104, USA
[17]CNRS, UMR 7095, Institut d'Astrophysique de Paris, F-75014, Paris, France
[18]Sorbonne Universités, UPMC Univ Paris 06, UMR 7095, Institut d'Astrophysique de Paris, F-75014, Paris, France
[19]Department of Physics, Northeastern University, Boston, MA 02115, USA
[20]University Observatory, LMU Faculty of Physics, Scheinerstr. 1, 81679 Munich, Germany
[21]Department of Physics & Astronomy, University College London, Gower Street, London, WC1E 6BT, UK
[22]Center for Astrophysics | Harvard & Smithsonian, 60 Garden Street, Cambridge, MA 02138, USA
[23]Kavli Institute for Particle Astrophysics & Cosmology, P. O. Box 2450, Stanford University, Stanford, CA 94305, USA
[24]SLAC National Accelerator Laboratory, Menlo Park, CA 94025, USA

[25]Brookhaven National Laboratory, Bldg 510, Upton, NY 11973, USA
[26]School of Mathematics and Physics, University of Queensland, Brisbane, QLD 4072, Australia
[27]Department of Physics, University of Genova and INFN, Via Dodecaneso 33, 16146, Genova, Italy
[28]Department of Physics, Carnegie Mellon University, Pittsburgh, Pennsylvania 15312, USA
[29]NSF AI Planning Institute for Physics of the Future,
Carnegie Mellon University, Pittsburgh, PA 15213, USA
[30]Instituto de Astrofisica de Canarias, E-38205 La Laguna, Tenerife, Spain
[31]Universidad de La Laguna, Dpto. Astrofísica, E-38206 La Laguna, Tenerife, Spain
[32]Institut de Física d'Altes Energies (IFAE), The Barcelona Institute of
Science and Technology, Campus UAB, 08193 Bellaterra (Barcelona) Spain
[33]Institut d'Estudis Espacials de Catalunya (IEEC), 08034 Barcelona, Spain
[34]Oxford College of Emory University, Oxford, GA 30054, USA
[35]Department of Astronomy and Astrophysics, University of Chicago, Chicago, IL 60637, USA
[36]Department of Physics, Duke University Durham, NC 27708, USA
[37]Instituto de Astrofísica de Canarias, Calle Vía Láctea s/n, E-38205, La Laguna, Tenerife, Spain
[38]Jodrell Bank Center for Astrophysics, School of Physics and Astronomy,
University of Manchester, Oxford Road, Manchester, M13 9PL, UK
[39]University of Nottingham, School of Physics and Astronomy, Nottingham NG7 2RD, UK
[40]Astronomy Unit, Department of Physics, University of Trieste, via Tiepolo 11, I-34131 Trieste, Italy
[41]Institute for Fundamental Physics of the Universe, Via Beirut 2, 34014 Trieste, Italy
[42]Hamburger Sternwarte, Universität Hamburg, Gojenbergsweg 112, 21029 Hamburg, Germany
[43]George P. and Cynthia Woods Mitchell Institute for Fundamental Physics and Astronomy,
and Department of Physics and Astronomy, Texas A&M University, College Station, TX 77843, USA
[44]Lawrence Berkeley National Laboratory, 1 Cyclotron Road, Berkeley, CA 94720, USA
[45]Department of Physics, IIT Hyderabad, Kandi, Telangana 502285, India
[46]Université Grenoble Alpes, CNRS, LPSC-IN2P3, 38000 Grenoble, France
[47]Department of Astronomy/Steward Observatory, University of Arizona,
933 North Cherry Avenue, Tucson, AZ 85721-0065, USA
[48]Jet Propulsion Laboratory, California Institute of Technology, 4800 Oak Grove Dr., Pasadena, CA 91109, USA
[49]Department of Physics and Astronomy, University of Waterloo,
200 University Ave W, Waterloo, ON N2L 3G1, Canada
[50]California Institute of Technology, 1200 East California Blvd, MC 249-17, Pasadena, CA 91125, USA
[51]Department of Astronomy, University of Michigan, Ann Arbor, MI 48109, USA
[52]Institute of Theoretical Astrophysics, University of Oslo. P.O. Box 1029 Blindern, NO-0315 Oslo, Norway
[53]Instituto de Física, UFRGS, Caixa Postal 15051, Porto Alegre, RS - 91501-970, Brazil
[54]Instituto de Física Teórica UAM/CSIC, Universidad Autónoma de Madrid, 28049 Madrid, Spain
[55]Department of Physics and Astronomy, Pevensey Building, University of Sussex, Brighton, BN1 9QH, UK
[56]Centre for Astrophysics & Supercomputing, Swinburne University of Technology, Victoria 3122, Australia
[57]Department of Astronomy, University of Illinois at Urbana-Champaign, 1002 W. Green Street, Urbana, IL 61801, USA
[58]School of Physics and Astronomy, Cardiff University, CF24 3AA, UK
[59]Department of Astronomy, University of Geneva, ch. d'Écogia 16, CH-1290 Versoix, Switzerland
[60]Santa Cruz Institute for Particle Physics, Santa Cruz, CA 95064, USA
[61]Center for Cosmology and Astro-Particle Physics, The Ohio State University, Columbus, OH 43210, USA
[62]Department of Physics, The Ohio State University, Columbus, OH 43210, USA
[63]MTA–CSFK Lendület "Momentum" Large-Scale Structure (LSS) Research Group,
Konkoly Thege Miklós út 15-17, H-1121 Budapest, Hungary
[64]Konkoly Observatory, HUN-REN Research Centre for Astronomy and Earth Sciences,
Konkoly Thege Miklós út 15-17, H-1121 Budapest, Hungary
[65]Institute of Cosmology and Gravitation, University of Portsmouth, Portsmouth, PO1 3FX, UK
[66]Department of Physics, University of Arizona, Tucson, AZ 85721, USA
[67]Australian Astronomical Optics, Macquarie University, North Ryde, NSW 2113, Australia
[68]Lowell Observatory, 1400 Mars Hill Rd, Flagstaff, AZ 86001, USA
[69]Department of Physics and Astronomy, Ohio University, Clippinger Labs, Athens, OH 45701
[70]Department of Astronomy and Astrophysics, University of Toronto,
50 St. George Street, Toronto ON, M5S 3H4, Canada
[71]Instituto de Astrofísica e Ciências do Espaço, Faculdade de Ciências, Universidade de Lisboa, 1769-016 Lisboa, Portugal
[72]Centre for Gravitational Astrophysics, College of Science,
The Australian National University, ACT 2601, Australia
[73]Department of Physics, Simon Fraser University, Burnaby, British Columbia, V5A 1S6, Canada
[74]Canadian Institute for Theoretical Astrophysics (CITA), University of Toronto,
60 St George Street, Toronto, Ontario M5S 3H8, Canada
[75]Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge CB3 0WA, UK
[76]Institute of Astronomy, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK
[77]Kavli Institute for Cosmology, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK

[78]Aix Marseille Univ, CNRS/IN2P3, CPPM, Marseille, France
[79]Institució Catalana de Recerca i Estudis Avançats, E-08010 Barcelona, Spain
[80]Department of Physics, University of Cincinnati, Cincinnati, Ohio 45221, USA
[81]Centro de Tecnologia da Informação Renato Archer, Campinas, SP, Brazil - 13069-901
[82]Observatório Nacional, Rua Gal. José Cristino 77, Rio de Janeiro, RJ - 20921-400, Brazil
[83]Perimeter Institute for Theoretical Physics, 31 Caroline St. North, Waterloo, ON N2L 2Y5, Canada
[84]Ruhr University Bochum, Faculty of Physics and Astronomy, Astronomical Institute, German Centre for Cosmological Lensing, 44780 Bochum, Germany
[85]University of Copenhagen, Dark Cosmology Centre, Juliane Maries Vej 30, 2100 Copenhagen O, Denmark
[86]ICRA, Centro Brasileiro de Pesquisas Físicas, Rua Dr. Xavier Sigaud 150, CEP 22290-180, Rio de Janeiro, RJ, Brazil
[87]ICTP South American Institute for Fundamental Research, IFT-UNESP, São Paulo, SP 01440-070, Brazil
[88]Laboratoire de physique des 2 infinis Irène Joliot-Curie, CNRS Université Paris-Saclay, Bât. 100, F-91405 Orsay Cedex, France
[89]McWilliams Center for Cosmology and Astrophysics, Department of Physics, Carnegie Mellon University, Pittsburgh, PA 15213, USA
[90]ICTP South American Institute for Fundamental Research Instituto de Física Teórica, Universidade Estadual Paulista, São Paulo, Brazil
[91]Departament de Física, Universitat Autònoma de Barcelona (UAB), 08193 Bellaterra, Barcelona, Spain
[92]Department of Physics, Stanford University, 382 Via Pueblo Mall, Stanford, CA 94305, USA
[93]Physics Department, Lancaster University, Lancaster, LA1 4YB, UK
[94]Computer Science and Mathematics Division, Oak Ridge National Laboratory, Oak Ridge, TN 37831
[95]School of Physics and Astronomy, University of Southampton, Southampton, SO17 1BJ, UK
[96]Central University of Kerala, Kasaragod, Kerala, India 671325
[97]Department of Physics, Oxford University, Oxford, UK
[98]Berkeley Center for Cosmological Physics, Department of Physics, University of California, Berkeley, CA 94720, US
[99]Max Planck Institute for Extraterrestrial Physics, Giessenbachstrasse, 85748 Garching, Germany
[100]Universitäts-Sternwarte, Fakultät für Physik, Ludwig-Maximilians Universität München, Scheinerstr. 1, 81679 München, Germany
[101]NSF's National Optical-Infrared Astronomy Research Laboratory, 950 N Cherry Avenue, Tucson, AZ, 85719
[102]Institute for Astronomy, University of Edinburgh, Edinburgh EH9 3HJ, UK

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We present results on dark energy evolution, assuming a time-dependent equation of state $w(a) = w_0 + w_a(1-a)$, from growth and geometric probes using the full six-year Dark Energy Survey (DES) dataset: type Ia supernovae (SNe), baryon acoustic oscillations (BAO), and weak gravitational lensing and galaxy clustering (3×2pt). The combination yields $w_0 = -0.84^{+0.10}_{-0.10}$ and $w_a = -0.44^{+0.60}_{-0.55}$, the tightest constraints ever obtained from a single survey, and shows a 2.2σ deviation from a cosmological constant. Adding the Dark Energy Spectroscopic Instrument (DESI) DR2 BAO data yields $w_0 = -0.84^{+0.06}_{-0.07}$ and $w_a = -0.53^{+0.33}_{-0.28}$, representing the most stringent low-redshift-only test of dynamical dark energy to date, with a 2.3σ deviation. In this combination, the inclusion of 3×2pt doubles the constraining power. Finally, when combined with primary CMB information from a combination of *Planck*, the Atacama Cosmology Telescope, and the South Pole Telescope, we obtain $w_0 = -0.82^{+0.05}_{-0.05}$, $w_a = -0.63^{+0.21}_{-0.18}$, 3.0σ away from a cosmological constant. We find that including 3×2pt in the previously studied SN + DESI BAO + CMB combination leaves the significance essentially unchanged (3.2σ to 3.0σ) while improving the figure of merit by ∼10%. We systematically investigate the impact of leaving out each one of the probes (SN, BAO, 3×2pt, CMB) and find that the significance of the deviation from a cosmological constant ranges from 2.3 to 3.2σ, with best-fit parameters always occupying the region where $w_0 > -1$ and $w_a < 0$. Excluding SN from the all data combination yields a 2.6σ departure from ΛCDM, providing a cross-check independent of supernova photometric calibration. We find the residual S_8 parameter difference between the CMB and 3×2pt datasets is unaffected by the assumption of dynamical dark energy. These results support the weak preference for evolving dark energy reported by several recent cosmological analyses. By combining growth and geometric probes from a single survey, this work realizes the multi-probe dark energy program envisioned at the inception of DES.

Keywords: dark energy; dark matter; cosmology: observations; cosmological parameters

INTRODUCTION

Understanding the mechanism driving the accelerated expansion of the Universe remains one of the most pressing open questions in modern cosmology. The cosmological constant, Λ, provides a simple and remarkably successful phenomenological description of dark energy, consistently fitting a wide range of cosmological observations over the past few decades [1–10]. However, its physical interpretation remains deeply problematic: if Λ is interpreted as the vacuum energy density of spacetime, its observed value is many orders of magnitude smaller than naive expectations from quantum field theory, giving rise to the well-known cosmological constant problem [11, 12]; if some physical principle sets the fundamental vacuum energy density to zero or to a value much smaller than the present energy density of the Universe, then it is plausible that dark energy arises as a dynamical phenomenon associated with novel fields not yet having reached their ground states [13].

A key question is therefore whether the dark energy density is truly a constant or instead evolves with time. Detecting any time dependence would provide crucial insight into the underlying nature of dark energy potentially pointing to new dynamical degrees of freedom. A wide class of theoretical models predict such evolution, including scalar field scenarios such as quintessence [13–15], k-essence [16], and phantom fields [17], each characterized by a time-dependent equation of state $w(a)$. The simplest parameterization is a constant w that is currently found to be consistent with -1 [3]. Going beyond that, a commonly used framework to capture such dynamics at low redshifts is the Chevallier–Polarski–Linder (CPL) parameterization [18, 19] (referred to as w_0w_aCDM):

$$w(a) = w_0 + w_a(1 - a)\,. \tag{1}$$

Recent observations of Type Ia supernovae (SNe) [4, 20] and Baryon Acoustic Oscillations (BAO) [21, 22] have provided intriguing hints of dynamical dark energy, which also persist when combined with Cosmic Microwave Background (CMB) data, with a significance at the 2–4σ level within the CPL framework [4, 5, 23–25]. Notably, the best-fit results in the context of the CPL parametrization suggest a transition in the equation of state parameter, with $w < -1$ at high redshift and $w > -1$ at low redshift. Such a "phantom-crossing" behavior [26, 27] cannot be realized within canonical single-field models and instead points toward more complex scenarios, such as non-minimal couplings between dark energy and dark matter [28, 29], or modifications of gravity [30–35]. In turn, these possibilities offer an avenue for probing the microphysics of dark energy.

The Dark Energy Survey (DES) [36] full six-year dataset (DES Y6) [37] enables an independent cross-check of such claims with unprecedented precision. In addition to BAO and SN data, which were used to explore constraints on dynamical dark energy in Ref. [25], here we also include the DES Y6 3×2pt measurements, which are based on a series of papers [3, 38–54]. When combined with BAO and SN, 3×2pt breaks degeneracies in parameter space and provides an independent assessment of dynamical dark energy without including CMB information. This is the most precise measurement to date of dark energy from a single survey. In addition, we also combine our DES-only results with external geometric and CMB constraints.

The full analysis details associated with this Letter, as well as additional extensions to the ΛCDM model, will be presented in an upcoming paper [55, hereafter Y6-EXT].

DATASETS

The Dark Energy Survey

Weak Gravitational Lensing and Galaxy Clustering: 3×2pt

The DES Y6 3×2pt analysis jointly measures cosmic shear, galaxy-galaxy lensing, and galaxy clustering [51, 54, 56] from two galaxy samples covering ∼4,300 $\deg^2$ [38]: a shear sample of ∼140 million galaxies in four tomographic bins [40, 42, 43, 50] up to $z \lesssim 2$ and a position sample of ∼9 million MAGLIM++ galaxies in six bins over $0.2 < z < 1.2$ [41, 44].

The analysis described in this Letter largely follows the methodology in Ref. [3, hereafter Y6-3×2PT], and we adopt the same cosmological parameter priors as in Ref. [Y6-3×2PT], except for the additional flat priors $w_0 \in [-3, -0.33]$ and $w_a \in [-3, 3]$, with $w_0 + w_a < 0$. The main differences are: (1) we define our analysis choices and scale cuts taking into account the additional constraining power from external data, (2) consequently we need to exclude additional small scales from cosmic shear, (3) we adopt the simpler nonlinear alignment (NLA) model for intrinsic alignments instead of the Tidal Alignment and Tidal Torque (TATT), and (4) we fix the sum of neutrino masses to $\sum m_\nu = 0.06\,\mathrm{eV}$, to reduce projection effects in the marginalized posteriors. The motivation for these changes is further discussed in Ref. [Y6-EXT], while in the next section we show that none of our results is driven by them. The blinding procedure follows that of Ref. [Y6-3×2PT]: cosmological results were hidden until the analysis choices and robustness tests were finalized for this model, at which point the chains were unblinded without further modifications.

Type Ia Supernovae

The DES supernova dataset comprises 1,623 photometrically identified type Ia supernovae (SNe) from the full DES five-year supernova survey [4, 57], combined with approximately 200 SNe from historical low-redshift samples [58–61]. In this work, we use DES-Dovekie [5], a reanalysis of the original DES SNe data with updated photometric cross-calibration. While the results obtained before and after recalibration are consistent with each other, when combined with the primary CMB [8–10] and the BAO measurements from DESI Data Release 2 [DR2, 23], the recalibrated sample yields a reduced significance for dynamical dark energy (3.2σ) compared to the original DES-SN5YR dataset (4.2σ) [5]. Other SN catalogues, namely Pantheon+ and Union3.1, also lead to similar dynamical dark energy significances (of 3.2σ and 3.4σ, respectively) after recent reanalyses, when combined with DESI DR2 BAO and different CMB combinations [7]. At the time of this analysis, a parallel effort is in progress combining the DES-Dovekie and the Pantheon+ datasets [62]. We leave the combination of that new SN dataset and other probes to future analyses.

Baryon Acoustic Oscillations

DES measures the baryon acoustic oscillation feature from the clustering of ∼16 million red and bright galaxies over 4,273 deg^2 in six tomographic bins in the redshift range $0.6 < z < 1.2$, at an effective redshift of $z_{\rm eff} = 0.851$ [22]. The combined measurement of angular-diameter distance to sound horizon scale of $D_M(z_{\rm eff})/r_d = 19.51 \pm 0.41$ represents a 2.1% constraint, the tightest BAO measurement from a photometric survey to date. Since the DES footprint overlaps with DESI by ∼1,000 deg^2, a separate analysis excluding this overlap region yields $D_M(z_{\rm eff})/r_d = 19.74 \pm 0.60$ [63], which we use when combining with DESI BAO to avoid double-counting information. When used without DESI BAO, the full DES BAO dataset is employed. Hereafter, we refer to the combination of DESI BAO and DES BAO (excluding the overlap region) simply as BAO.

External Data

To further constrain a dynamic dark energy, we also combine DES data with state-of-the-art measurements from other experiments.

We use DESI DR2 BAO [23]. The DESI data consist of ratios of distances, either parallel or transverse to the line-of-sight, to the sound horizon at the baryon drag epoch, r_d. These distances are measured from the two-point correlation function of several tracers of the matter distribution, including galaxies, quasars and the Lyman-α forest. The tracers' effective redshifts cover the range $0.295 < z < 2.330$.

Additionally, we consider CMB temperature and polarization (TT, EE and TE) anisotropy power spectra from the *Planck* collaboration Public Release 3 (PR3) [8], the Atacama Cosmology Telescope DR6 [9], and South Pole Telescope SPT-3G DR1 [10]. These datasets are combined as described in Ref. [Y6-3×2PT], and the combination has been previously considered in Ref. [10]. We do not include CMB lensing since a proper combination would require modeling the cross-covariance with the 3×2pt probe.

RESULTS

In this section, we present constraints on the w_0–w_a parameter space. We use the Nautilus nested sampler [64] for all parameter inference reported here. Unless otherwise stated, all parameter constraints are quoted as the posterior mean in each parameter with the 68% credible level (C.L.) of posterior volume. In Fig. 1, we show the posteriors in the w_0–w_a parameter plane for multiple dataset combinations, including 3×2pt. In Table I, we report the distance to the cosmological constant case in units of σ, which we define based on the fraction of posterior mass enclosed by the contour that intersects the $(w_0, w_a) = (-1, 0)$ point, following the posterior-difference method in Ref. [Y6-3×2PT]. We also show, for each combination, its corresponding dark energy figure of merit, FoM $\equiv 1/\sqrt{\det(\mathrm{Cov}_{w_0 w_a})}$, which measures the overall strength of the constraints in the w_0–w_a plane [65]. We also report the $\Delta\chi^2$-based significance, with $\Delta\chi^2 \equiv \chi^2_{w_0 w_a \rm CDM} - \chi^2_{\Lambda \rm CDM}$ computed at the maximum-a-posteriori (MAP) point of each model, which closely matches the one reported by the posterior-difference method. We have also checked that the evidence ratios corresponding to these data combinations show a picture consistent with the metrics listed in Table I.

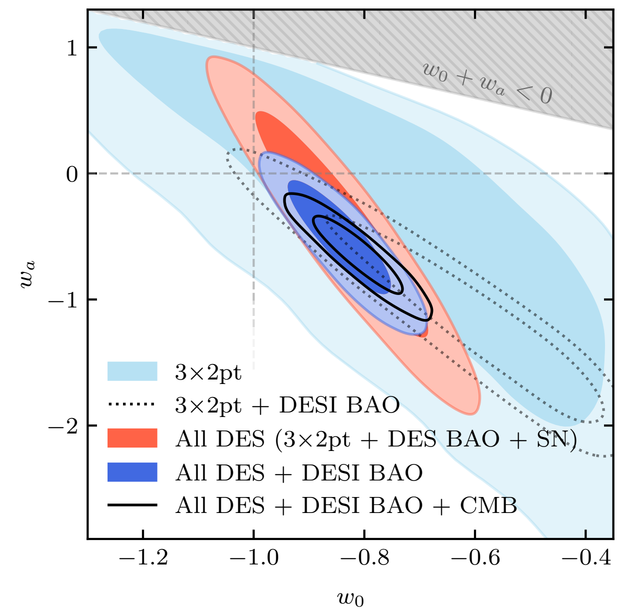

FIG. 1. Constraints on the $w_0 w_a$CDM parameters from DES 3×2pt data (light blue), from DES 3×2pt data and DESI BAO (dotted), from all DES probes (3×2pt + SN + DES BAO; red filled), from all DES probes combined with DESI BAO (3×2pt + SN + DES BAO + DESI BAO; blue filled), from all datasets considered in this work combined (3×2pt + SN + BAO + CMB; black solid). The dashed crosshair marks the cosmological constant values ($w_0 = -1$, $w_a = 0$). The hatched region is excluded by the prior $w_0 + w_a < 0$. All contours show 68% and 95% credible regions.

The DES 3×2pt data are consistent with a cosmological constant at the 1.0σ level. Marginalized constraints (95% C.L.) yield $w_0 > -1.26$ and $w_a = -0.43^{+1.10}_{-0.78}$, representing an improvement of 22% over DES Year 3 on w_a [66]. For the first time, we combine 3×2pt with DES BAO and SN measurements to provide a measurement of w_0 and w_a parameters that uses only cosmological probes from a single survey. We obtain:

$$\left.\begin{aligned} w_0 &= -0.84 \pm 0.10 \\ w_a &= -0.44^{+0.60}_{-0.55} \end{aligned}\right\} \begin{array}{l}\text{All DES (68\% C.L.)} \\ (3\times2\text{pt + DES BAO + SN}),\end{array} \quad (2)$$

a result that is 2.2σ away from the cosmological constant value $(w_0, w_a) = (-1, 0)$. In this combination, 3×2pt data improves the FoM by a factor of four compared to the purely geometric DES BAO + SN, showing the complementarity between those datasets.

When further combining our data with BAO measurements from DESI, we obtain:

$$\left.\begin{aligned} w_0 &= -0.84^{+0.06}_{-0.07} \\ w_a &= -0.53^{+0.33}_{-0.28} \end{aligned}\right\} \begin{array}{l}\text{All DES + DESI BAO} \\ \text{(68\% C.L.)},\end{array} \quad (3)$$

which is 2.3σ away from ΛCDM. This result is still entirely based on low-redshift data, independent of the CMB. As can be seen in Table I, including 3×2pt in this combination increases the FoM from 61 to 110, highlighting the importance of the 3×2pt information for low-redshift tests of dark energy. Adding CMB data to the low-redshift combination roughly doubles the FoM (from 110 to 222), owing to the long distance lever arm to recombination. With the combination All DES + DESI BAO + CMB we obtain:

$$\left.\begin{aligned} w_0 &= -0.82 \pm 0.05 \\ w_a &= -0.63^{+0.21}_{-0.18} \end{aligned}\right\} \begin{array}{l}\text{All DES + DESI BAO + CMB} \\ \text{(68\% C.L.)},\end{array} \quad (4)$$

which is 3.0σ away from ΛCDM. Compared to the SN + BAO + CMB combination, the addition of 3×2pt improves the FoM from 202 to 222 (10%), while the change in the ΛCDM-departure significance (3.2σ to 3.0σ) is consistent with the agreement we find between our two significance metrics (see Table I).

In the lower part of Table I, we investigate the robustness of our results by systematically removing one probe type at a time from the full dataset combination, and report the distance to ΛCDM and the FoM for each new combination, similar to the study in Ref. [67]. In each case, the remaining data show varying levels of difference from ΛCDM, ranging from 2.3σ (excluding CMB) to 3.2σ (excluding 3×2pt). These variations are stable in terms of their central values, with best-fit parameters always occupying the region of $w_0 > -1$ and $w_a < 0$. Thanks to the improved constraining power of the DES Y6 3×2pt, we are also able to perform an inference leaving out *both* SN and CMB. The combination of 3×2pt + DESI BAO deviates from ΛCDM at 1.7σ, providing a test of dynamical dark energy independent of SN and CMB.

In Fig. 2, we summarize the best-constrained values of the pivot w_p from the various dataset combinations explored in this work. The pivot value is defined as $w_p \equiv w(z_p)$, where the pivot redshift z_p is chosen such that $\mathrm{Cov}(w_p, w_a) = 0$. The DES data, including 3×2pt, allow us to have multiple redundant combinations of probes that can constrain w_p at a range of pivot redshifts spanning much of the late-time expansion history. Without the DES data, only the single CMB + DESI BAO point (rightmost) would be available.

TABLE I. Distance to ΛCDM in the w_0w_aCDM parameter space for different dataset combinations, defined as the fraction of posterior mass enclosed by the isolikelihood contour that intersects $(w_0, w_a) = (-1, 0)$. We also show $\Delta\chi^2$-based model comparison, defined as $\Delta\chi^2 \equiv \chi^2_{w_0w_a\mathrm{CDM}} - \chi^2_{\Lambda\mathrm{CDM}}$ computed at the maximum-a-posteriori (MAP) point of each model. Significances from both metrics are reported in units of Gaussian standard deviations, following Ref. [Y6-3×2PT]. The last column shows the dark energy figure of merit FoM $\equiv 1/\sqrt{\det(\mathrm{Cov}_{w_0w_a})}$. We refer to the combination of DESI BAO and DES BAO (excluding the area overlap, see text for details) as BAO, while SN corresponds to the DES Dovekie recalibrated sample.

Datasets	Distance to ΛCDM (σ)	$\Delta\chi^2$ (σ)	FoM
3×2pt	1.0	1.0	6
SN + DES BAO	1.5	1.6	11
All DES (3×2pt + SN + DES BAO)	2.2	2.0	48
All data (All DES + DESI BAO + CMB)	3.0	3.0	222
SN + BAO	2.4	2.3	61
3×2pt + DESI BAO	1.7	1.5	32
CMB + DESI BAO	2.9	2.9	61
All DES + DESI BAO (All except CMB)	2.3	2.2	110
SN + BAO + CMB (All except 3×2pt)	3.2	3.3	202
3×2pt + BAO + CMB (All except SN)	2.6	2.5	66
3×2pt + SN + CMB (All except BAO)	2.4	2.4	96
All DES + CMB (All except DESI BAO)	2.8	2.8	102

We have also verified that allowing $\sum m_\nu$ to vary (assuming a uniform prior of $\sum m_\nu > 0.06\,\mathrm{eV}$) does not significantly shift the w_0–w_a constraints; the best-fit values and significance levels remain consistent with the fixed-mass case. Our tightest combination yields an upper bound of $\sum m_\nu < 0.18$ eV (95% CL) with best-fit values $w_0 = -0.80^{+0.06}_{-0.06}$ and $w_a = -0.74^{+0.25}_{-0.21}$, yielding a distance to a cosmological constant of 3.4σ. The slight increase from 3.0σ to 3.4σ reflects a shift in the central value of w_a toward more negative values driven by the $\sum m_\nu$–w_a degeneracy.

Finally, we test the robustness of the low-redshift-only case (All DES + DESI BAO) and the combination of all data to alternative analysis choices, including different intrinsic alignment models (TATT and NLA with free amplitude per redshift bin), freeing the baryonic feedback amplitude, and including the second tomographic bin of the position sample excluded from the baseline (see Ref. [Y6-EXT] for details). For each variant, we compute both the 1D shift in w_0, w_a, Ω_m, and S_8 relative to the baseline, and also the change in the significance of the difference of the fit from ΛCDM. We find that all parameter shifts are within 0.1σ in w_0 and w_a, within 0.2σ for Ω_m, and S_8, and that the distance to ΛCDM changes by no more than $\pm0.3\sigma$. These tests confirm that our results are not sensitive to these analysis choices. We note that the second tomographic bin of the position sample, excluded from our baseline following Ref. [Y6-3×2PT], continues to

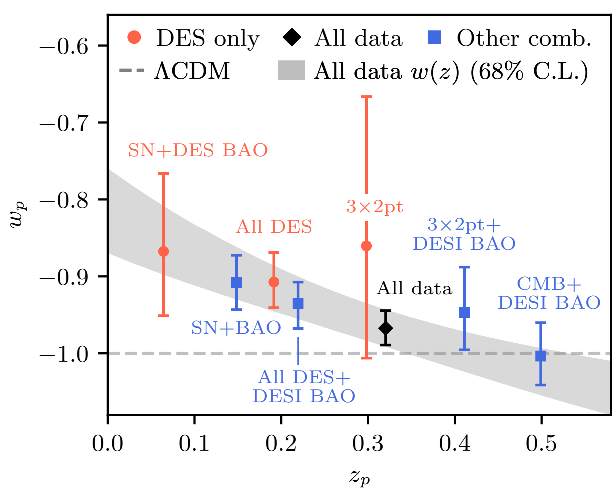

FIG. 2. The best-constrained values of the pivot w_p are shown at their pivot redshift for a variety of combinations of probes studied in this paper. The 68% C.L. of $w(z)$ obtained using the CPL parametrization is shown as the shaded region for the combination of all data. The DES data, now including 3×2pt, allow us to have multiple redundant combinations of probes that can constrain w_p at a range of pivot redshifts spanning much of the late-time expansion history.

show redshift uncertainty values lying outside its calibration prior when included, confirming that this issue persists in the w_0w_aCDM model.

Beyond the dark energy constraints, we also examine the implication for S_8 in the context of evolving dark energy. We find the S_8 difference between 3×2pt and the primary CMB unchanged. The full analysis is presented in the Supplemental Material.

CONCLUSION

In this Letter, we present constraints on dynamical dark energy from a combination of growth and geometric probes from the full six-year dataset from DES. These are the first constraints on the evolution of dark energy using multiple probes from a *single* Stage III survey — a major milestone for the experimental dark energy program. This result combines one growth probe (3×2pt) and two geometry probes (SN and BAO) from DES, achieving a dark energy FoM of 48. The marginalized values yield $w_0 = -0.84^{+0.10}_{-0.10}$, $w_a = -0.44^{+0.60}_{-0.55}$, and is 2.2$\sigma$ away from a cosmological constant.

Going beyond a single survey and combining the DES-only results with DESI DR2 BAO data yields the most stringent low-redshift-only test of dynamical dark energy to date. We find $w_0 = -0.84^{+0.06}_{-0.07}$ and $w_a = -0.53^{+0.33}_{-0.28}$. The inclusion of 3×2pt in this combination doubles the constraining power. The constraining power doubles again when combining these datasets with primary CMB data. The final constraints are $w_0 = -0.82^{+0.05}_{-0.05}$, $w_a = -0.63^{+0.21}_{-0.18}$, 3.0$\sigma$ away from a cosmological constant.

The redundancy of growth and geometry probes in our final combination also allows us to examine the sensitivity of our results when excluding one individual measurement at a time from the full combination. In these tests the posterior contours remain stable in location with best-fit parameters consistently in the region $w_0 > -1$, $w_a < 0$, and the significance varies from 2.3σ to 3.2σ. The pivot values of w from these redundant combinations of probes span a broad redshift range ($z_p \sim$ 0.05–0.5), underscoring that these constraints probe the dark energy equation of state across much of the late-time expansion history.

This Letter marks a significant step towards completing the multi-probe dark energy program laid out in the early 2000s [68], setting the stage for the next chapter of the program with Stage-IV imaging surveys such as Rubin, Euclid, and Roman.

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AUTHORS' CONTRIBUTION

- Weak lensing and galaxy clustering analysis (3×2pt) [Y6-3×2PT]: M. Adamow, A. Alarcón, A. Amon, D. Anbajagane, W. d'Assignies, S. Avila, J. Beas-Gonzalez, K. Bechtol, M. R. Becker, G. M. Bernstein, J. Blazek, G. Camacho-Ciurana, G. Campailla, A. Campos, A. Carnero Rosell, C. Chang, J. M. Coloma-Nadal, M. Crocce, J. DeRose, T. Diehl, A. Drlica-Wagner, J. Elvin-Poole, S. Everett, A. Ferté, M. Gatti, G. Giannini, R. A. Gruendl, W. G. Hartley, M. Jarvis, E. Krause, S. Lee, E. Legnani, S. Mau, J. Mena-Fernández, F. Menanteau, J. Muir, J. Myles, A. Pieres, A. Porredon, J. Prat, M. Raveri, W. Riquelme, M. Rodríguez-Monroy, P. Rogozenski, A. Roodman, E. S. Rykoff, S. Samuroff, C. Sanchez, D. Sanchez-Cid, T. Schutt, I. Sevilla-Noarbe, E. Sheldon, M. Tabbutt, C.-H. To, M. A. Troxel, J. de Vicente, N. Weaverdyck, A. Whyley, M. Yamamoto, B. Yanny, B. Yin.

- BAO analysis [22]: M. Adamow, S. Avila, K. Bechtol, H. Camacho, R. Cawthon, K. C. Chan, L. Toribio San Cipriano, M. Crocce, T. M. Davis, T. Diehl, A. Drlica-Wagner, J. Elvin-Poole, I. Ferrero, G. Giannini, R. A. Gruendl, W. G. Hartley, J. Mena-Fernández, W. J. Percival, A. Pieres, A. Porredon, M. Rodriguez-Monroy, A. Carnero Rosell, A. J. Ross, E. S. Rykoff, C. Sanchez, E. Sanchez, I. Sevilla-Noarbe, E. Sheldon, J. de Vicente, N. Weaverdyck, B. Yanny.

- Supernova Type Ia analysis [4]: P. Armstrong, D. Brout, R. Camilleri, A. Carr, R. Chen, T. M. Davis, J. Frieman, L. Galbany, K. Glazebrook, S. R. Hinton, R. Kessler, J. Lee, C. Lidman, A. Möller, B. Popovic, H. Qu, M. Sako, D. Scolnic, P. Shah, M. Smith, M. Sullivan, B. O. Sánchez, G. Taylor, M. Toy, M. Vincenzi, P. Wiseman.

- Analysis development and pipeline: O. Alves, G. Campailla, A. Ferté, K. Koyama, S. Lee, A. Liddle, M. X. Lin, J. Muir, M. Raveri, J. Rebouças, P. Rogozenski, R. Rosenfeld, D. Sanchez-Cid, N. Shah, and D. Souza.

- Parameter estimation, validation, and robustness tests: O. Alves, G. Campailla, S. Lee, J. Muir, M. Raveri, and J. Rebouças.

- Paper writing: O. Alves, M. R. Becker, G. Campailla, C. Chang, M. Crocce, A. Ferté, J. Frieman, K. Koyama, S. Lee, A. Liddle, M. X. Lin, J. Muir, J. Prat, M. Raveri, J. Rebouças, M. A. Troxel.

- Coordination and scientific management: M. R. Becker and M. Crocce (Year Six Key Project Coordinators), C. Chang and M. A. Troxel (Science Committee Chairs), A. Alarcón, A. Amon, O. Alves, S. Avila, J. Blazek, A. Ferté, G. Giannini, S. Lee, A. Porredon, J. Prat, M. Raveri, M. Rodríguez-Monroy, C. Sanchez, D. Sanchez-Cid, T. Schutt, N. Weaverdyck, M. Yamamoto, B. Yin (Working Group and Analysis Team leads).

The remaining authors have made contributions to this paper that include, but are not limited to, the construction of DECam and other aspects of collecting the data; data processing and calibration; developing broadly used methods, codes, and simulations; running the pipelines and validation tests; and promoting the science analysis.

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Supplemental Material

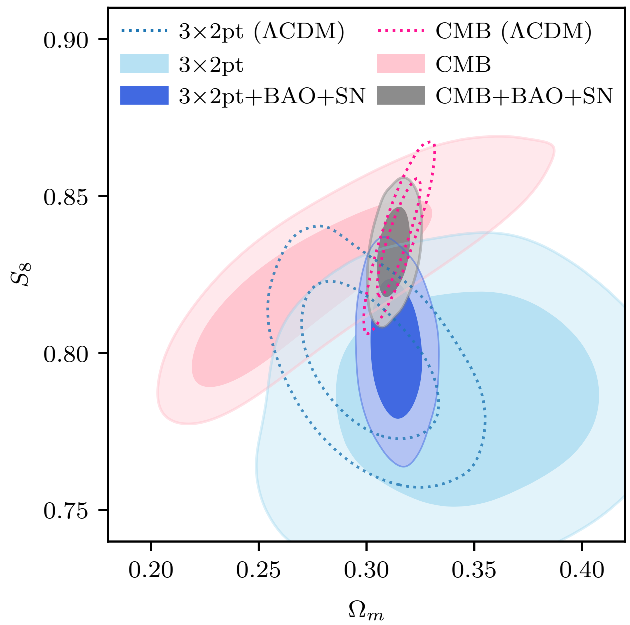

FIG. S1. Constraints on Ω_m and S_8 under ΛCDM and w_0w_aCDM. Dotted contours show ΛCDM constraints from the CMB (pink) and 3×2pt (light blue). Filled contours show the corresponding w_0–w_a results: CMB alone (pink filled) and 3×2pt alone (light blue filled) exhibit large degeneracies, and the formal tension is reduced. Adding BAO + SN to anchor the expansion history yields CMB + BAO + SN (gray) and 3×2pt + BAO + SN (dark blue).

Implications for Growth of Structure

The DES Y6 3×2pt ΛCDM results provide some of the tightest constraints on growth of structure in the low-redshift universe [Y6-3×2PT]. Comparing the inferred S_8 to high-redshift predictions from the primary CMB is a powerful test of the concordance cosmology model. In Ref. [Y6-3×2PT], we found that 3×2pt and the primary CMB have a parameter difference at the 2.2σ level in the Ω_m–S_8 plane, and 2.6σ in the S_8 direction. With our baseline analysis choices and assuming a ΛCDM model, we obtain 1.9σ and 2.0σ, respectively. The differences relative to values reported in Ref. [Y6-3×2PT] reflect our different analysis choices described above, and are within the range validated by our robustness tests [S55].

There have been many similar tests in the literature that have found 1–3σ differences [S1, S2, S71–S81] – this is often discussed under the term 'S_8 tension.' Here, we are interested in whether the overall picture changes when we move to a dynamical dark energy model.

Figure S1 shows the Ω_m–S_8 constraint from 3×2pt and the primary CMB under ΛCDM and w_0w_aCDM. Since the w_0w_aCDM model has more degrees of freedom in the late-time expansion history, the constraining power on Ω_m and S_8 with only CMB or 3×2pt is reduced. The larger error bars reduce the offset to 0.4σ in S_8 and 1.1σ in the Ω_m–S_8 plane. The constraints on the growth parameters are

$$\left.\begin{aligned} \Omega_\mathrm{m} &= 0.32 \pm 0.03 \\ S_8 &= 0.81 \pm 0.02 \end{aligned}\right\} \text{3×2pt (68\% C.L.)}, \tag{S1}$$

consistent with the ΛCDM result as shown in Fig. S1.

We further examine this difference when additional geometric anchors from BAO and SN are included. The geometric data provide tighter constraints on Ω_m and break the degeneracy between S_8 and Ω_m. Under this common anchoring, we compare constraints from CMB + BAO + SN and 3×2pt + BAO + SN. The two combinations show parameter difference at 1.2σ in the Ω_m–S_8 plane and 1.7σ in S_8 (see Table S1 in the Supplemental Material for full parameter constraints). We note that in ΛCDM, the same geometric anchoring results in a 0.7σ difference in the Ω_m–S_8 plane and 1.1σ in S_8. This suggests that the overall consistency between the growth parameters in the early and late Universe remain unchanged under w_0w_aCDM.

Marginalized parameter constraints

In Table S1 we summarize the marginalized parameter constraints and MAP values obtained from the various dataset combinations explored in this paper in the w_0w_aCDM model.

TABLE S1. Summary of marginalized parameter constraints and MAP values (in-between parenthesis) in the w_0w_aCDM model.

Dataset	S_8	Ω_m	σ_8	$\Omega_\mathrm{b}\times 10^2$	n_s	h	w_0	w_a	w_p	z_p
3×2pt	$0.810^{+0.023}_{-0.022}$ (0.834)	0.324 ± 0.031 (0.316)	$0.782^{+0.036}_{-0.040}$ (0.813)	$5.1^{+1.1}_{-1.0}$ (5.6)	$0.963^{+0.015}_{-0.030}$ (0.932)	$0.681^{+0.051}_{-0.081}$ (0.773)	< -1.3 95% CL	$-0.4^{+1.1}_{-0.8}$ (-0.0)	$-0.86^{+0.19}_{-0.14}$ (-0.72)	0.30
SN + DES BAO	–	$0.320^{+0.065}_{-0.041}$ (0.400)	–	$4.2^{+0.3}_{-1.2}$ (3.0)	$0.965^{+0.019}_{-0.029}$ (0.948)	$0.714^{+0.084}_{-0.031}$ (0.800)	$-0.80^{+0.12}_{-0.11}$ (-0.75)	$-1.1^{+1.2}_{-0.9}$ (-3.0)	$-0.87^{+0.10}_{-0.08}$ (-0.93)	0.06
All DES (3 × 2pt + SN + DES BAO)	$0.809^{+0.016}_{-0.018}$ (0.816)	$0.301^{+0.010}_{-0.012}$ (0.295)	$0.808^{+0.027}_{-0.026}$ (0.822)	$4.62^{+0.85}_{-0.69}$ (5.27)	$0.964^{+0.017}_{-0.029}$ (0.937)	$0.708^{+0.071}_{-0.047}$ (0.791)	-0.84 ± 0.10 (-0.82)	$-0.44^{+0.60}_{-0.55}$ (-0.44)	-0.91 ± 0.04 (-0.90)	0.19
All data (All DES + DESI BAO + CMB)	0.823 ± 0.008 (0.824)	0.311 ± 0.005 (0.312)	0.808 ± 0.008 (0.808)	$4.949^{+0.079}_{-0.083}$ (4.956)	0.974 ± 0.003 (0.974)	$0.674^{+0.006}_{-0.005}$ (0.674)	-0.82 ± 0.05 (-0.78)	$-0.63^{+0.21}_{-0.18}$ (-0.75)	-0.97 ± 0.02 (-0.97)	0.32
SN + BAO	$0.92^{+0.32}_{-0.27}$ (1.18)	$0.312^{+0.017}_{-0.012}$ (0.314)	$0.90^{+0.31}_{-0.24}$ (1.16)	$5.0^{+1.0}_{-0.7}$ (5.3)	$0.965^{+0.019}_{-0.028}$ (0.943)	$0.689^{+0.060}_{-0.082}$ (0.713)	$-0.84^{+0.06}_{-0.08}$ (-0.84)	$-0.54^{+0.44}_{-0.48}$ (-0.55)	$-0.91^{+0.03}_{-0.04}$ (-0.91)	0.15
3×2pt + DESI BAO	$0.808^{+0.016}_{-0.015}$ (0.809)	$0.337^{+0.022}_{-0.015}$ (0.356)	$0.763^{+0.022}_{-0.026}$ (0.743)	$5.27^{+0.85}_{-0.80}$ (5.03)	$0.961^{+0.013}_{-0.029}$ (0.938)	$0.662^{+0.031}_{-0.076}$ (0.626)	$-0.61^{+0.24}_{-0.11}$ (-0.44)	$-1.17^{+0.47}_{-0.72}$ (-1.78)	$-0.95^{+0.06}_{-0.05}$ (-0.96)	0.41
CMB + DESI BAO	$0.846^{+0.012}_{-0.011}$ (0.851)	$0.341^{+0.018}_{-0.011}$ (0.352)	$0.794^{+0.012}_{-0.014}$ (0.786)	$5.37^{+0.27}_{-0.18}$ (5.53)	0.971 ± 0.003 (0.971)	$0.647^{+0.009}_{-0.017}$ (0.637)	$-0.54^{+0.19}_{-0.07}$ (-0.43)	$-1.40^{+0.29}_{-0.50}$ (-1.67)	-1.00 ± 0.04 (-0.99)	0.50
All DES + DESI BAO (All except CMB)	$0.801^{+0.015}_{-0.014}$ (0.796)	0.313 ± 0.008 (0.310)	$0.785^{+0.020}_{-0.019}$ (0.782)	$4.9^{+1.0}_{-0.6}$ (5.0)	$0.965^{+0.021}_{-0.024}$ (0.967)	$0.687^{+0.059}_{-0.067}$ (0.686)	$-0.84^{+0.06}_{-0.07}$ (-0.87)	$-0.53^{+0.33}_{-0.28}$ (-0.43)	-0.94 ± 0.03 (-0.95)	0.22
SN + BAO + CMB (All except 3 × 2pt)	$0.833^{+0.010}_{-0.009}$ (0.832)	$0.313^{+0.005}_{-0.006}$ (0.311)	$0.816^{+0.010}_{-0.009}$ (0.817)	$4.937^{+0.072}_{-0.083}$ (4.919)	0.972 ± 0.003 (0.972)	$0.675^{+0.006}_{-0.005}$ (0.676)	$-0.80^{+0.05}_{-0.06}$ (-0.81)	$-0.72^{+0.22}_{-0.20}$ (-0.72)	-0.97 ± 0.02 (-0.97)	0.30
3×2pt + BAO + CMB (All except SN)	$0.834^{+0.010}_{-0.009}$ (0.829)	$0.337^{+0.018}_{-0.012}$ (0.330)	$0.788^{+0.012}_{-0.015}$ (0.791)	$5.34^{+0.28}_{-0.20}$ (5.25)	0.973 ± 0.003 (0.974)	$0.650^{+0.010}_{-0.017}$ (0.655)	$-0.57^{+0.20}_{-0.10}$ (-0.62)	$-1.25^{+0.33}_{-0.51}$ (-1.10)	-1.00 ± 0.04 (-1.00)	0.53
3×2pt + SN + CMB (All except BAO)	0.823 ± 0.008 (0.820)	0.305 ± 0.007 (0.302)	$0.816^{+0.011}_{-0.010}$ (0.817)	$4.84^{+0.10}_{-0.12}$ (4.81)	0.973 ± 0.003 (0.974)	0.682 ± 0.008 (0.684)	$-0.76^{+0.08}_{-0.09}$ (-0.80)	$-0.95^{+0.42}_{-0.38}$ (-0.81)	$-0.96^{+0.02}_{-0.03}$ (-0.96)	0.25
All DES + CMB (All except DESI BAO)	$0.822^{+0.007}_{-0.008}$ (0.821)	0.303 ± 0.007 (0.301)	$0.819^{+0.010}_{-0.009}$ (0.820)	$4.81^{+0.10}_{-0.11}$ (4.79)	0.973 ± 0.003 (0.974)	$0.684^{+0.007}_{-0.008}$ (0.686)	-0.74 ± 0.08 (-0.72)	$-1.10^{+0.42}_{-0.36}$ (-1.17)	$-0.96^{+0.03}_{-0.02}$ (-0.95)	0.25